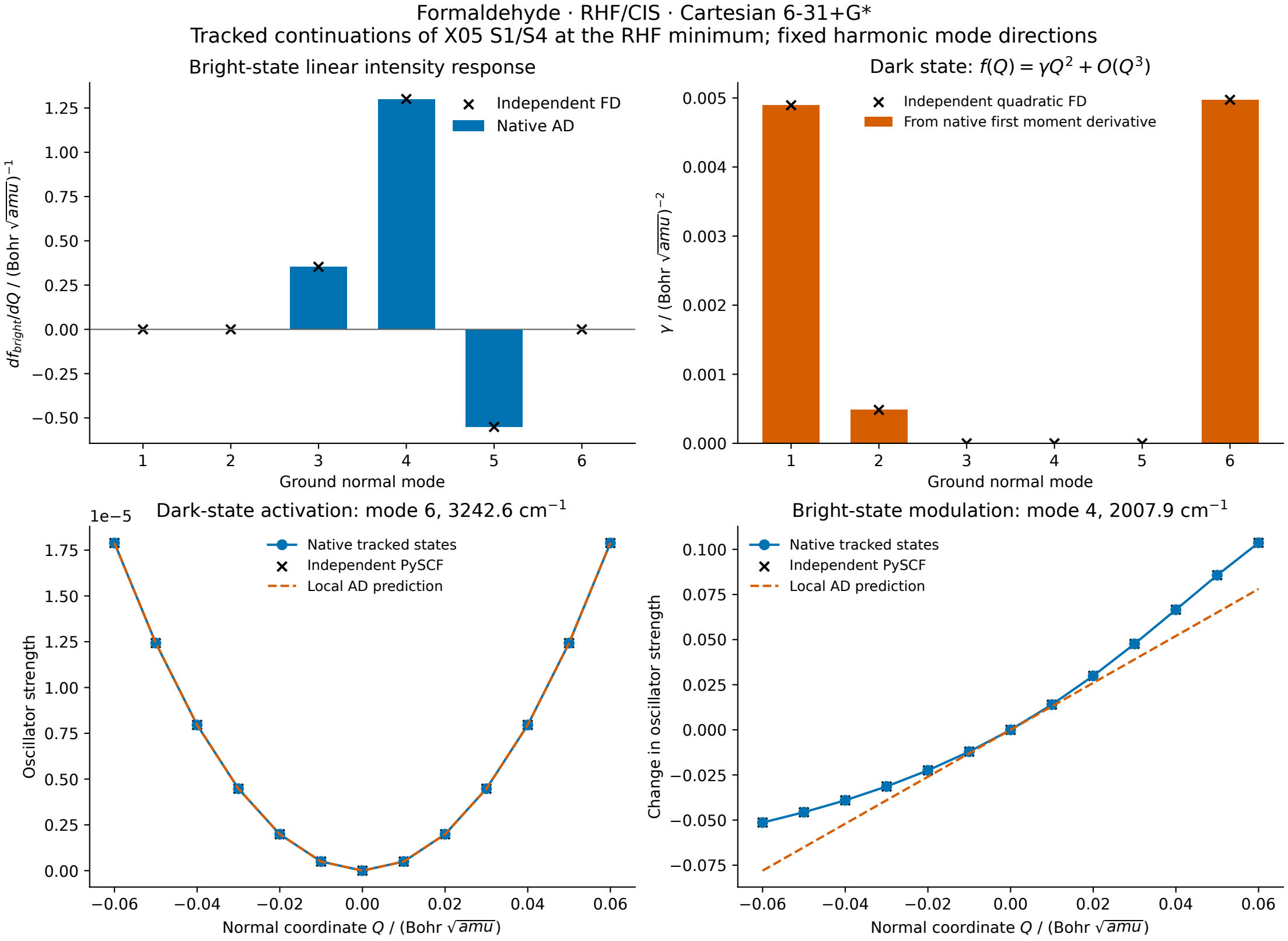
Formaldehyde · RHF/CIS · Cartesian 6-31+G*
Tracked continuations of X05 S1/S4 at the RHF minimum; fixed harmonic mode directions
Bright-state linear intensity response
Independent FD
Native AD
df_{bright}/dQ / (Bohr $\sqrt{amu}$)$^{-1}$
Ground normal mode
Dark state: $f(Q) = \gamma Q^2 + O(Q^3)$
Independent quadratic FD
From native first moment derivative
γ / (Bohr $\sqrt{amu}$)$^{-2}$
Ground normal mode
Dark-state activation: mode 6, 3242.6 cm^{-1}
1e−5
Native tracked states
Independent PySCF
Local AD prediction
Oscillator strength
Normal coordinate Q / (Bohr $\sqrt{amu}$)
Bright-state modulation: mode 4, 2007.9 cm^{-1}
Native tracked states
Independent PySCF
Local AD prediction
Change in oscillator strength
Normal coordinate Q / (Bohr $\sqrt{amu}$)